MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 101: MECHANICS

Date: 23rd February, 2018 Time: 3.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE in Section A and any other TWO in Section B.

SECTION A

QUESTION 1 (30 MARKS)

a) (i) State Newton's second law of motion and apply the law to derive the Newton's equation of motion.

(ii) Starting with Newton's equation of motion, prove that the kinetic energy of a free body is conserved.

b) For a body of mass m under an external force of the form $F = \alpha t^2 + \beta t$, where α, β are constants, determine: (i) the acceleration (ii) the speed (iii) total energy at any time t, taking the initial speed to be V_0.

c) (i) Write down Newton's force of gravity on a body of mass m at a distance r from the centre of the earth, given that the mass of the earth is M_E.

(ii) Determine the gravitational potential energy of the body specified in (c) (i).

SECTION B

Answer any TWO questions in this section.

QUESTION 2 (20 MARKS)

a) (i) State and prove Kepler's second law of planetary motion.

(ii) A body of mass 50 tonnes is launched from the surface of the earth into outer space.

Calculate the minimum kinetic energy which must be given to the body for a successful launch. Assume the earth to be a sphere of radius $R = 6370\ km$ and the acceleration due to gravity to be $g = 9.81\ ms^{-2}$.

b) Consider a body subjected to a frictional force proportional to its speed v in the form $F = -\alpha V$, where α is a constant. Determine:

(i) the impulsive linear momentum of the body.

(ii) the kinetic energy of the body.

QUESTION 3 (20 MARKS)

a) A linear harmonic oscillator of mass m and angular frequency ω has potential energy $U = \frac{1}{2}m\omega^2 x^2$ where x is the displacement along the axis.

(i) Calculate the force on the oscillator.

(ii) Calculate the linear momentum of the oscillator.

b) If the oscillator in (a) is driven by an external time varying force of the form $F = A\cos(\Omega t)$, A, Ω are constants, calculate the total energy transferred to the body due to the action of the external force.

QUESTION 4 (20 MARKS)

a) (i) A body starting off within initial kinetic energy T_0 is driven by an external force $F = \sin t + t^3 - t^2$. Determine the kinetic energy of the body at any time t.

(ii) A projectile starting off with initial speed $25ms^{-1}$ moves vertically upwards. If the acceleration due to gravity is $g = 9.81ms^{-2}$, determine the highest position it can reach.

b) (i) Show that under the action of a constant external force F, the speed of a body of mass m is related to the displacement x in the form

$V^2 = V_0^2 + \frac{2F}{m}(x - x_0)$ where V_0 is the initial speed and x_0 is the initial position.

(ii) A ball of mass $10g$ moving with speed $20ms^{-1}$ collides directly with another ball of mass $5g$ moving with speed $15ms^{-1}$ in the opposite direction. If the two bodies have respective speeds $25ms^{-1}$ and $18ms^{-1}$, determine their coefficient of restitution.

QUESTION 5 (20 MARKS)

A body of mass m subject to displacement and time varying forces is governed by Newton's equation of motion of the form $\frac{d^2x}{dt^2} = -kx + a\sin\beta t$, where k, a, β are constants.

Determine:

(i) the general solution describing time varying displacement $x(t)$.

(ii) the kinetic energy of the body at any time t

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UNIVERSITY EXAMINATIONS 2017/2018

FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN INDUSTRIAL CHEMISTRY WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 104: THERMAL PHYSICS

Date: 13th February, 2018 Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO from SECTION B.

SPH 104: THERMAL PHYSICS

<u>**Instructions**</u>

Section A is Compulsory

Answer ANY TWO Questions from Section B

<u>**Useful Constants**</u>

Specific heat of water, 4190 J/kg·K

Specific heat capacity of ice, 2090 J/kg·K

Specific heat capacity of steam, 2010 J/kg·K

Heat of fusion of water, 334×10^3 J/kg

Heat of vaporization of water, 2256×10^3 J/kg

Stefan's constant, 5.67×10^{-8} W/m^2·K^4

Universal gas constant, 8.31 J/mol·K

Section A

Question One [30 mks]

a) Explain the following terms;

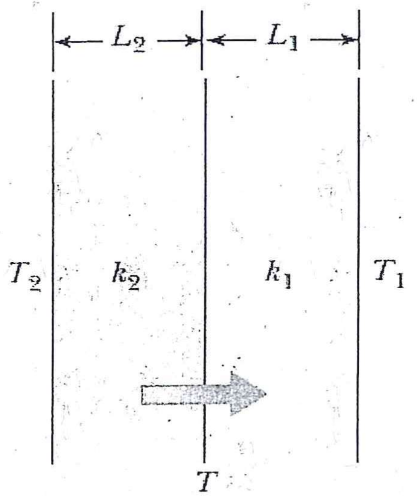
i) Heat **[2 mks]**

ii) Temperature **[2 mks]**

iii) Thermal equilibrium **[2 mks]**

iv) Zeroth law of thermodynamics **[2 mks]**

b) Two slabs of thickness L_1 and L_2 and thermal conductivities k_1 and k_2 are in thermal contact with each other as shown.

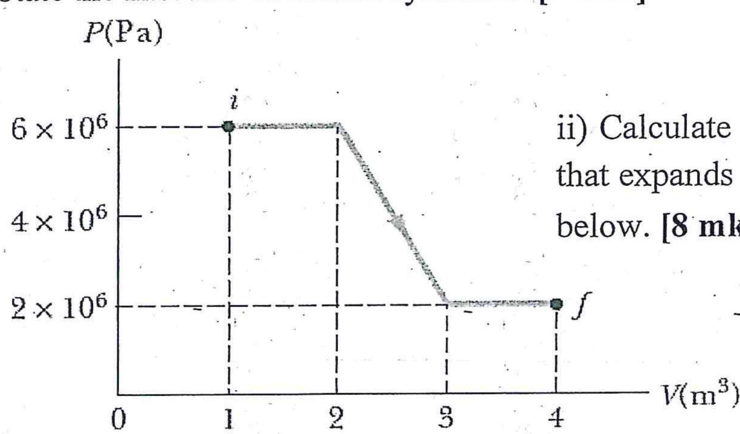


The temperatures of their outer surfaces are T_1 and T_2, respectively, and $T_2>T_1$. Obtain expressions for:

i) The temperature at the interface. **[6 mks]**

ii) The rate of energy transfer by conduction through the slabs in the steady-state condition. **[6 mks]**

c) i) State the first law of thermodynamics. **[2 mks]**



ii) Calculate the work done by a fluid that expands from i to f in the figure below. **[8 mks]**

Section B:

Question Two [20 mks]

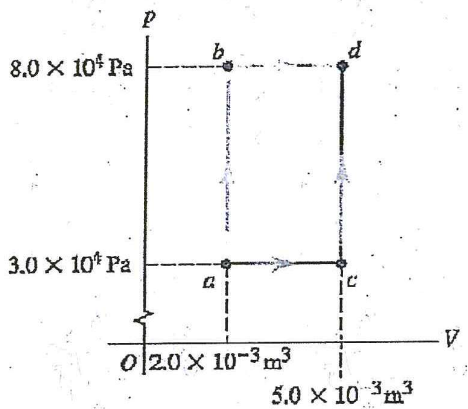
a) Explain the following types of thermodynamic processes.

i) Adiabatic **[2 mks]**

ii) Isobaric **[2 mks]**

iii) Isochoric **[2 mks]**

b) A series of thermodynamic processes is shown in the P-V diagram bellow. In process ab, 150 J of heat is added to the system and in process bd, 600 J of heat is added. Find:



(i) The internal energy change in process *ab*. **[3 mks]**

(ii) The internal energy change in process *abd*. **[4 mks]**

(iii) The total heat added in process *acd*. **[3 mks]**

c) A sample of ideal gas is expanded to twice its original volume of $1.00\ m^3$ in a quasi-static process for which $P = \alpha V^2$. Obtain an expression for the work done by the expanding gas. **[4 mks]**

Question Three [20 mks]

a) For a monatomic ideal gas, show that:

i) $C_v = \frac{3}{2}R$ **[5 mks]**

ii) $C_p = C_v + R$ **[5 mks]**

where C_p and C_v are the molar heat capacities at constant pressure and volume, respectively, and R is the universal gas constant.

b) Show that for an ideal gas undergoing an adiabatic expansion, TV^{γ} is a constant where $\gamma = C_p/C_v$. **[10 mks]**

Question Four [20 mks]

a) Define the following terms.

i) Latent heat **[1 mks]**

ii) Specific heat capacity **[1 mks]**

b) Calculate the heat energy required to convert 1.0 g of ice at -30°C to steam at 130°C. **[12 mks]**

c) A 0.05 ingot of a metal is heated to 200°C and then dropped into a beaker containing 0.4 kg of water initially at 20°C. If the final equilibrium temperature of the mixed system is 22.4°C, find the specific heat of the metal. **[6 mks]**

Question Five [20 mks]

a) i) Explain radiation as mode of heat transfer. **[2 mks]**

ii) The tungsten filament of a certain 100-W light bulb radiates 2.00 W of light. The filament has a surface area of 0.250 mm^2 and an emissivity of 0.950. Find the filament's temperature. **[5 mks]**

b) i) Explain what you understand by the term black body. **[2 mks]**

ii) The surface of the Sun has a temperature of about $5800\ K$. The radius of the Sun is $6.96 \times 10^8\ m$. Estimate the total energy radiated by the Sun each second. **[6 mks]**

iii) At high noon, the sun delivers 1000 W to each square meter of a blacktop road. If the hot asphalt loses energy only by radiation, what is its equilibrium temperature? **[5 mks]**.

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SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 201: DYNAMICS

Date: 23rd February, 2018 Time: 3.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE in Section A and any other TWO in Section B.

SECTION A

Q1. a) State the conditions for equilibrium (2MK

b) Figure 1.1 shows a handwinch used to move a load of 3000 N. What is the minimum effort (Force) required to turn the drum? (6MK

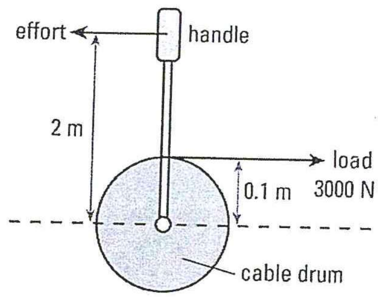

Figure 1.1

c) A wheel starting from rest, rotates with a constant angular acceleration of 2.0 rad/s^2. During a certain 3.0 s interval it turns through 90 rad. (a) What was the angular velocity of the wheel at the start of the 3.0 s interval ? (b) How long had the wheel been turning before the start of the 3.0 s interval ? (6Mks

d) State the principle of moments. (2MKs

e) Show that for a rod of length L and mass m and of uniform mass distribution the moment of inertia with respect to an axis through its center of mass is given by $I = \frac{1}{12}mL^2$. See Figure 1.2 below. (8MKs

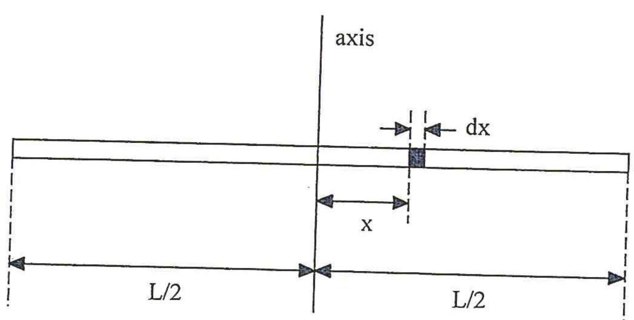

Figure 1.2

f) A 12.0 kg toy train car moving at 2.40 m/s on a straight, level train track, collides head-on with a second train car whose mass is 36.0 kg and was at rest on the track. If the collision is perfectly elastic and all motion is frictionless, calculate the velocities of the two cars after the collision. (6MKs)

SECTION B

Q2. a) Show that the total work done on a rotating rigid body is equal to (4Mks)

to the change in its rotational kinetic energy

b) Consider the rod in Figure 1.1. What is the moment of inertia with respect to an axis through its left corner? (4MKs)

c) Show that for a body of mass M that is rolling without slipping, the total kinetic energy is given by: (6Mks)

$$K = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

where v_{cm} is the velocity of the centre of mass and I_{cm} is the moment of inertia of the body about an axis through the centre of mass.

d) i) Derive an expression for power associated with work done by a torque acting on a rotating rigid body. (4MKs)

ii) An electric motor consumes 9 kJ of electrical energy in 1 min. If one third of this energy goes into heat and other forms of internal energy of the motor, with the rest going to the moor output, how much torque will this engine develop if you run it at 2500 rpm? (2MKs)

Q3. a) A small solid marble of mass m and radius r rolls without slipping along a loop-the-loop track shown in Figure 3.1, having been released from rest somewhere along the straight section of the track. From what minimum height above the bottom of the track must the marble be released in order not to leave the track at the top of the loop. (6Mks)

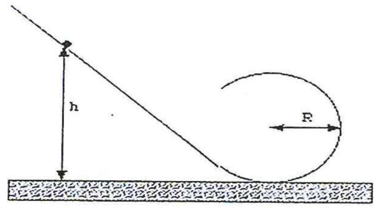

Figure 3.1

b) The rotational inertia of a collapsing spinning star changes to one-third of its initial value. What is the ratio of the new rotational kinetic energy to the initial rotational kinetic energy ? (8MKs

c) A cockroach with mass m runs counterclockwise around the rim of a lazy Susan (a circular dish mounted on a vertical axle) of radius R and rotational inertia I with frictionless bearings, see Figure 3.2. The cockroach's speed (with respect to the earth) is v, whereas the lazy Susan turns clockwise with angular speed θ_o. The cockroach finds a bread crumb on the rim and, of course, stops. (a) What is the angular speed of the lazy Susan after the cockroach stops ? (b) Is mechanical energy conserved ? (6MKs

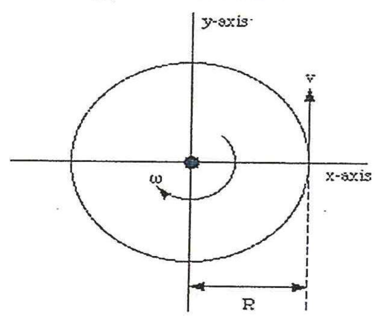

Figure 3.2

Q4. a) Define the term torque (3Mks)

b) i) Show that the work done by a net external torque, τ, acting on on a rotating body is equal to the change in kinetic energy. (6MKs)

ii) Show that the rate, P, at which a rotating body does work is (6MKs)

$$P = \tau\omega$$

c) A 1.5 kg grinding wheel is in the form of a solid cylinder of radius 0.10 m. What constant torque will bring it from rest to an angular speed of 1200 rev/min in 4 s? (5MKs)

Q5. a) What are the differences between inertial and non inertial frames of reference? **(5MKs)**

b) In Galilean transformation, we assume that $\Delta t = \Delta t'$. Also we assume that the origins coincide at $t = 0$. Write the transformation equations from the S to the S' frame for displacement, velocity and acceleration in the $x -$ coordiante. Note that for two inertial frames, the acceleration, $a_x = a_x'$. **(10MKs)**

c) What are the postulates of special relativety? **(5MKs)**

End

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UNIVERSITY EXAMINATIONS 2017/2018

SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 202: ELECTRICITY AND MAGNETISM II

Date: 16th February, 2018 Time: 8.30 -11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO from SECTION B

SPH 202: ELECTRICITY AND MAGNETISM II

Instructions:

- **Section A is compulsory**
- **Answer any two questions from section B**

Useful constants:

$\varepsilon_0 = 8.854\times10^{-12}\,C^2/Nm^2$

$e = -1.60\times10^{-19}\,C$

$m_p = 1.67\times10^{-27}\,Kg$

Section A

Question One [30 mks]

a) i) Give the mathematical statement of Gauss's law, defining all the quantities. **[2 mks]**

ii) What is a Gaussian surface? **[2 mks]**

b) An insulating solid sphere of radius a has a uniform volume charge density ρ and carries a total positive charge Q. Calculate the magnitude of the electric field at a point:

i) Outside the sphere. **[4 mks]**

ii) Inside the sphere. **[6 mks]**

c) i) Two parallel metallic plates of equal area A are separated by a distance d. One plate carries a charge Q and the other a charge $-Q$. Show that the capacitance C of the arrangement is proportional to A and inversely proportional to d. **[6 mks]**

ii) A parallel-plate capacitor has an area of $A = 2.00\times10^{-4}\,m^2$ and a plate separation $d = 1.00\,mm$. Find its capacitance. **[4 mks]**

d) A particle with charge $-1.24\times10^{-8}\,C$ is moving with instantaneous velocity v=(4.19×10^{4}m/s)i+(-3.85×10^{4}m/s)j. What is the magnitude of the force on this particle by a magnetic field B=(1.4T)i where i is a unit vector in the x-direction. **[6 mks]**

Section B

Question Two [20 mks]

a) i) Consider the magnetic force on a charged particle moving with a constant velocity, **v**, in a uniform magnetic field, **B,** perpendicular to the velocity. If the particle has charge, q, obtain an expression for its cyclotron frequency. **[7 mks]**

ii) A proton is moving in a circular orbit of radius 14 cm in a uniform 0.35 T magnetic field perpendicular to the velocity of the proton. Find the cyclotron frequency of the proton. **[6 mks]**

b) In the Borh model of the hydrogen atom in the lowest energy state, the electron orbits the proton. Show that the magnitude of its magnetic dipole moment, μ, is given by

$$\mu = \frac{e}{2m_e} L \, ,$$

where e is the electronic charge, m_e is the electronic mass and L is the magnitude of its angular momentum. **[7 mks]**

Question Three [20 mks]

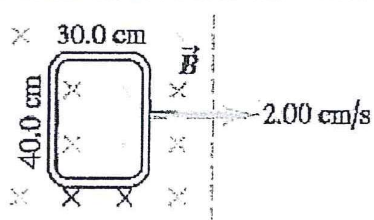
a) State Faraday's law of electromagnetic induction. **[4 mks]**

b) A flat, rectangular coil consisting of 50 turns measures 25 cm by 30 cm. It is in a uniform 1.2 T magnetic field, with the plane of the coil parallel to the field. In 0.22s it is rotated so that the plane of the coil is perpendicular to the field.

i) What is the change in the magnetic flux through the coil due to this rotation? **[5 mks]**

ii) Find the magnitude of the average emf induced in the coil due to this rotation. **[5 mks]**

c) A rectangular coil measuring 30.0 cm by 40 cm is located inside a uniform magnetic field of 1.25 T with the field perpendicular to the plane of the coil as shown in the figure below. The coil is pulled out at a steady rate of 2.0 cm/s travelling perpendicular to the field lines. Find the emf induced in the coil. **[6 mks]**



Question Four [20 mks]

Using Gauss's law, derive expressions for:

i) The electric field due to a positive point charge, q. **[5 mks]**

ii) An infinite plane sheet of charge of uniform positive surface charge density, σ. **[7 mks]**

iii) A line charge of uniform positive linear charge density, λ. **[8 mks]**

Question Five [20 mks]

a) Show that the torque $\vec{\tau}$ on an electric dipole moment $\vec{p}$ in an external electric field $\vec{E}$ is given by the cross product:

$$\vec{\tau} = \vec{p} \times \vec{E} \quad \textbf{[6 mks]}$$

b) i)Show that the potential energy U an electric dipole moment $\vec{p}$ in an external electric field $\vec{E}$ is given by the scalar:

$$U = -\vec{p} \bullet \vec{E} \quad \textbf{[8 mks]}$$

ii) The water molecule has an electric dipole moment of $6.3 \times 10^{-30}\, C \cdot m$. The dipole moment is oriented in the direction of an electric field of magnitude $2.5 \times 10^{5}\, N \cdot m$. Calculate the amount of work required to rotate the dipole from this orientation to one in which the dipole moment is perpendicular to the field. **[6 mks]**

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SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 203: MATHEMATICAL METHODS FOR PHYSICS I

Date: 12th February, 2018 Time: 8.30 -11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO from SECTION B

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UNIVERSITY EXAMINATIONS FOR 2017/2018 ACADEMIC YEAR

SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY (MAIN CAMPUS)

SPH 203: MATHEMATICAL METHODS FOR PHYSICS I

SEPTEMBER-DECEMBER, 2017 SESSION

INSTRUCTION

Answer question *1 (Compulsory)* and any other *TWO* questions in **section B.**

SECTION A

QUESTION 1 (30 MARKS)

a) Distinguish between a sequence and a series. **(2 marks)**

b) Prove that the sequence $\{a_n\} = \frac{1}{n}$ is Cauchy. **(3 marks)**

c) Determine the limit of the sequence $\ln\left(\frac{4n+1}{3n-1}\right)$ **(2 marks)**

d) Find the radius of convergence for the power series defined by $\sum_{k=1}^{\infty} k!(x+10)^k$ **(3 marks)**

e) The partition function of a spin-$\frac{1}{2}$ paramagnetic system is given by the exponential function

$Z = e^{\left(\frac{g_s\mu_B B}{2k_B T}\right)} + e^{-\left(\frac{g_s\mu_B B}{2k_B T}\right)}$. Express the equation as a hyperbolic function. **(2 marks)**

f) Obtain the derivative of the function $y = \sin 5x + \ln x^3 - e^{4x} + \sum_{k=0}^{\infty} c_k x^k$ **(4 marks)**

g) By using the method of integration by parts, evaluate $\int_0^N \ln x dx$ to obtain the Stirling's approximation for a system of N particles. **(3 marks)**

h) State the antiderivative form of the fundamental theorem of Calculus. **(1 mark)**

i) A physical system traces a path defined by the following parametric equations during its motion. $x = 2t^3 + 1$ and $y = t^2 \cos t$. Determine the gradient function of the path. **(3 marks)**

j) Calculate the work done when an object is pushed by a force $\vec{F} = (10\hat{i} + 15\hat{j})N$ through a displacement $\vec{s} = (3\hat{i} + 4\hat{j})m$. (2 marks)

k) Define the following terms.

(i) Linear independence. (1 mark)

(ii) Basis. (1 mark)

(iii) Asymptotic analysis (1 mark)

l) The unitless equivalents of Pauli spin matrices are given by $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$;

$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ and $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ where $i = \sqrt{-1}$. Show that the commutation bracket

$[\sigma_x, \sigma_y] = 2i\sigma_z$ (2 marks)

SECTION B

Answer any TWO questions in this section.

QUESTION 2 (20 MARKS)

a)Determine $\lim_{x\to\infty}\left(\dfrac{6x^2+5x+7}{4x^2+2x}\right)$ using de l'H$\hat{o}$pital's rule. (4 marks)

b) Derive the expression for the vector product $\vec{A}\times\vec{B}$ where $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$;

$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ (6 marks)

c) Given $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 3\hat{i} + 2\hat{j} + \hat{k}$. Find:

(i) $\vec{A}X\vec{B}$ (3 marks)

(ii) $\vec{A}.\vec{B}$ (2 marks)

(iii) Angle between $\vec{A}$ and $\vec{B}$ (2 marks)

(iv) Unit vector perpendicular to both $\vec{A}$ and $\vec{B}$. (3 marks)

QUESTION 3 (20 MARKS)

a) Apply the concept of integration of rational functions to evaluate

$$\int \frac{x^2+2x+4}{(x+1)^3}dx$$ (6 marks)

b) A particle described by the wave function $u = \frac{1}{\sqrt{a}}\cos\left(\frac{\pi x}{2a}\right)$ moves in a one-dimensional potential defined by $V = V_0 \cos\left(\frac{\pi x}{2a}\right)$. Calculate the first order energy correction given by

$E_1 = \int_{-a}^{a} uVudx$ **(14 marks)**

QUESTION 4 (20 MARKS)

a) Expand $f(x) = \sin x$ in a Taylor series about $\frac{\pi}{3}$ **(6 marks)**

b) Show that the series $\sum_{k=1}^{\infty} ar^{k-1}$ converges to $\frac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$ **(6 marks)**

c) A ladder $20\ m$ leans against a vertical wall. The foot of the ladder is being drawn away from the wall at the rate of $2\ m/s$. How fast is the top of the ladder sliding down at the instant when the foot is $10\ m$ from the wall? **(8 marks)**

QUESTION 5 (20 MARKS)

a) Use the substitution method to evaluate $\int 3\cos 3t dt$ **(4 marks)**

b) The primitive translation vectors of the body centred cubic (bcc) crystal lattice are given by

$\vec{a} = \frac{1}{2}a(\hat{x} + \hat{y} - \hat{z})$; $\vec{a} = \frac{1}{2}a(-\hat{x} + \hat{y} - \hat{z})$; $\vec{a} = \frac{1}{2}a(\hat{x} - \hat{y} + \hat{z})$. Calculate the volume of the primitive cell using the formula $V_{bcc} = |\vec{a}.\vec{b} \times \vec{c}|$. **(8 marks)**

c) (i) Define a linear map. **(2 marks)**

(ii)Let a map T be defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$. Show that T is linear. **(6 marks)**